COMSATS UNIVERSITY ISLAMABAD

Attock Campus

Department Of Computer Science

MUHAMMAD SUFYAN | FA24-BSE-049

Information Security LAB

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Assignment :04

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TASK 1: Simple RSA Implementation

Code:

```
# Function to find GCD
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a

# Function to find modular inverse
def mod_inverse(e, phi):
    for d in range(1, phi):
        if (e * d) % phi == 1:
            return d
    return None

# Step 1: Choose two prime numbers
p = 61
q = 53

# Step 2: Calculate n
n = p * q

# Step 3: Calculate phi(n)
phi = (p - 1) * (q - 1)

# Step 4: Choose e
e = 17  # must be coprime with phi

# Step 5: Find d
d = mod_inverse(e, phi)

print("Public Key:", (e, n))
print("Private Key:", (d, n))

# Encryption
def encrypt(message):
    return [(ord(char) ** e) % n for char in message]

# Decryption
def decrypt(cipher):
    return ''.join([chr((char ** d) % n) for char in cipher])

# Test
```

```
message = "Sufyan"
cipher = encrypt(message)
print("Encrypted:", cipher)

decrypted = decrypt(cipher)
print("Decrypted:", decrypted)
```

OUTPUT:

```
Active code page: 65001

D:\informationSecutiy>"C:\Program Files\Python313\python.exe" d:/informationSecutiy/informationSecurity_ta
sk/IS_LAB_MID.py/is_lab3.py/is_lab4.py/task1.py
Public Key: (17, 3233)
Private Key: (2753, 3233)
Encrypted: [2680, 2160, 1369, 487, 1632, 2235]
Decrypted: Sufyan

D:\informationSecutiy>
```

TASK 2: Digital Signatures (RSA + SHA 256)

Code:

```
import hashlib

# Hash function
def hash_message(message):
    return hashlib.sha256(message.encode()).hexdigest()

# Sign message
def sign(message, d, n):
    hashed = int(hash_message(message), 16)
    signature = pow(hashed, d, n)
    return signature

# Verify signature
```

```python
def verify(message, signature, e, n):
    hashed = int(hash_message(message), 16)
    decrypted_hash = pow(signature, e, n)
    return hashed == decrypted_hash

# Keys from previous RSA
message = "Hello World"

signature = sign(message, d, n)
print("Signature:", signature)

# Verification (correct message)
print("Verification:", verify(message, signature, e, n))

# Modify message
fake_message = "Hello world"
print("Verification after change:", verify(fake_message, signature, e, n))
```

OUTPUT:

```
D:\informationSecutiy>"C:\Program Files\Python313\python.exe" d:/informationSecutiy/informationSecurity_ta
sk/IS_LAB_MID.py/is_lab3.py/is_lab4.py/task2.py
Signature: 2622
Verification: False
Verification after change: False

D:\informationSecutiy>
```